AUTOMATED MELANOMA DETECTION USING DEEP LEARNING

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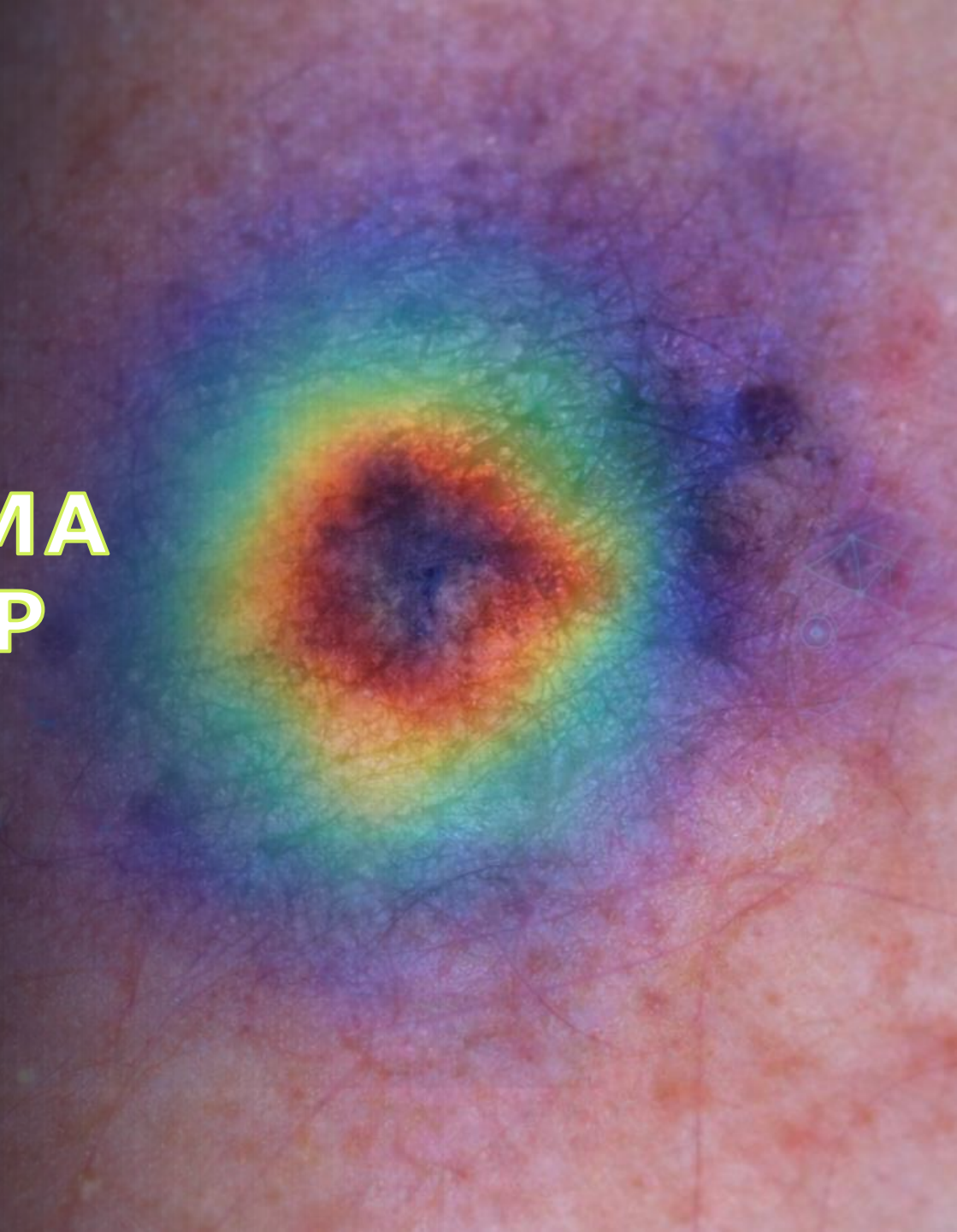
PhD Interview



ABOUT MELANOMA

What is Melanoma?

- A malignant skin cancer arising from melanocytes
- Responsible for the majority of skin cancer deaths

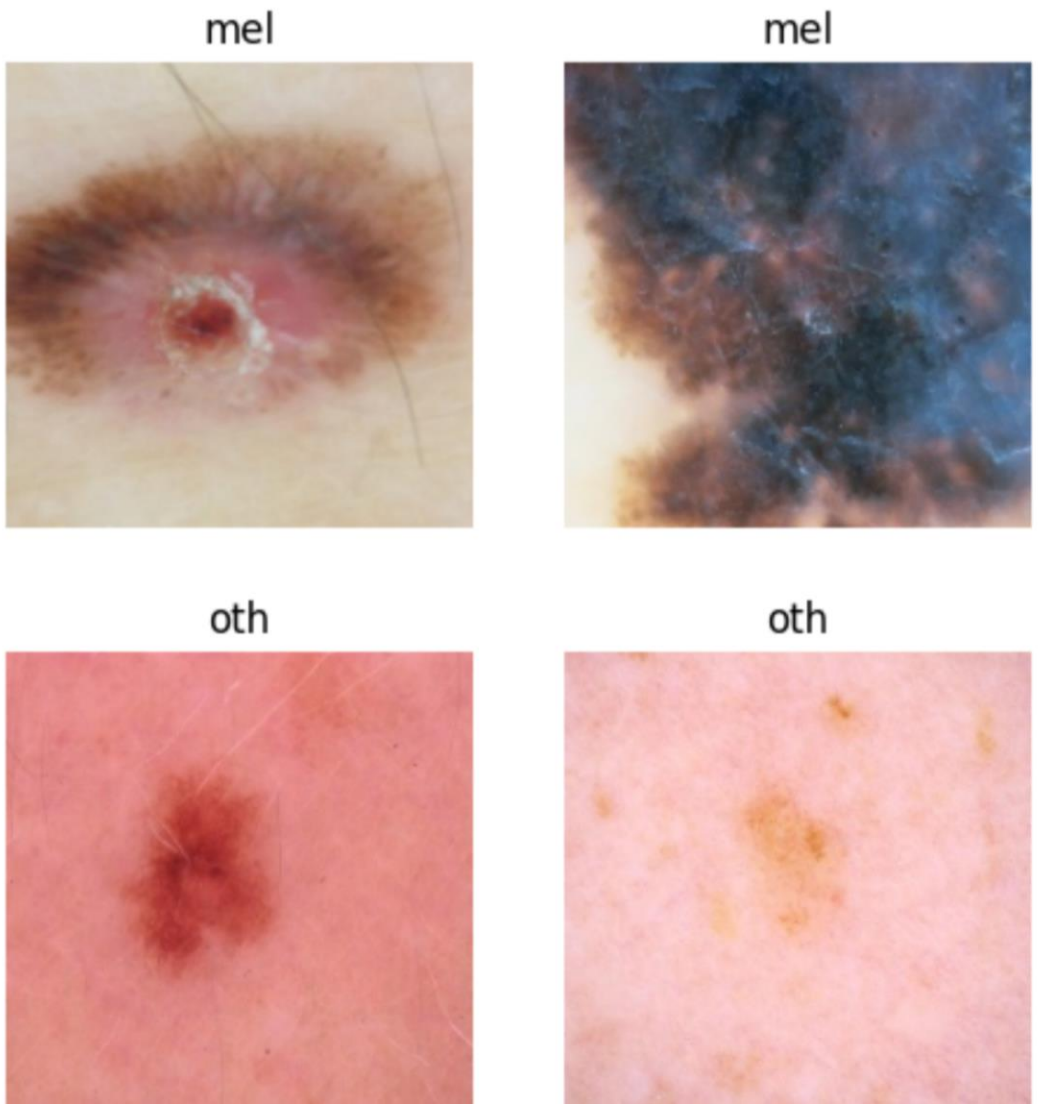
Why it matters:

- Early detection significantly improves survival rates
- Visual diagnosis is difficult due to variability in appearance

Clinical challenge:

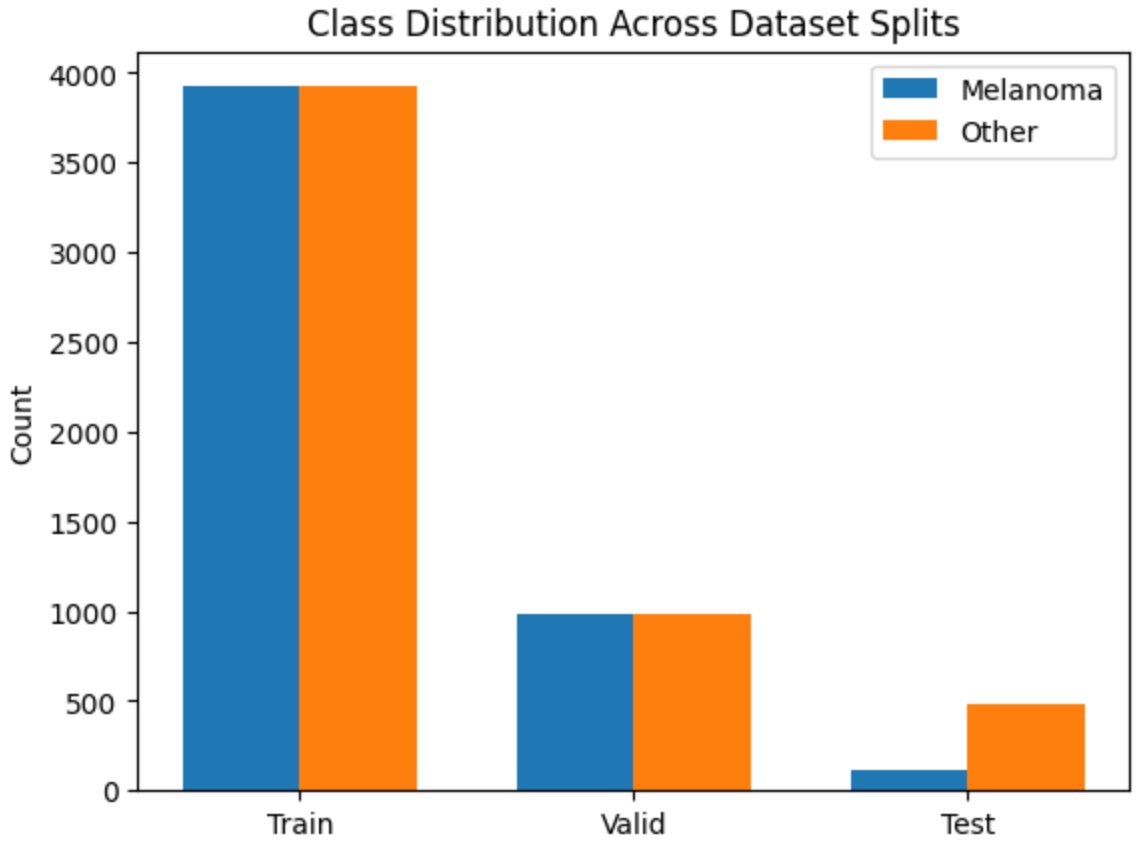
- Lesions vary in size, colour, and texture
- Often confused with benign lesions

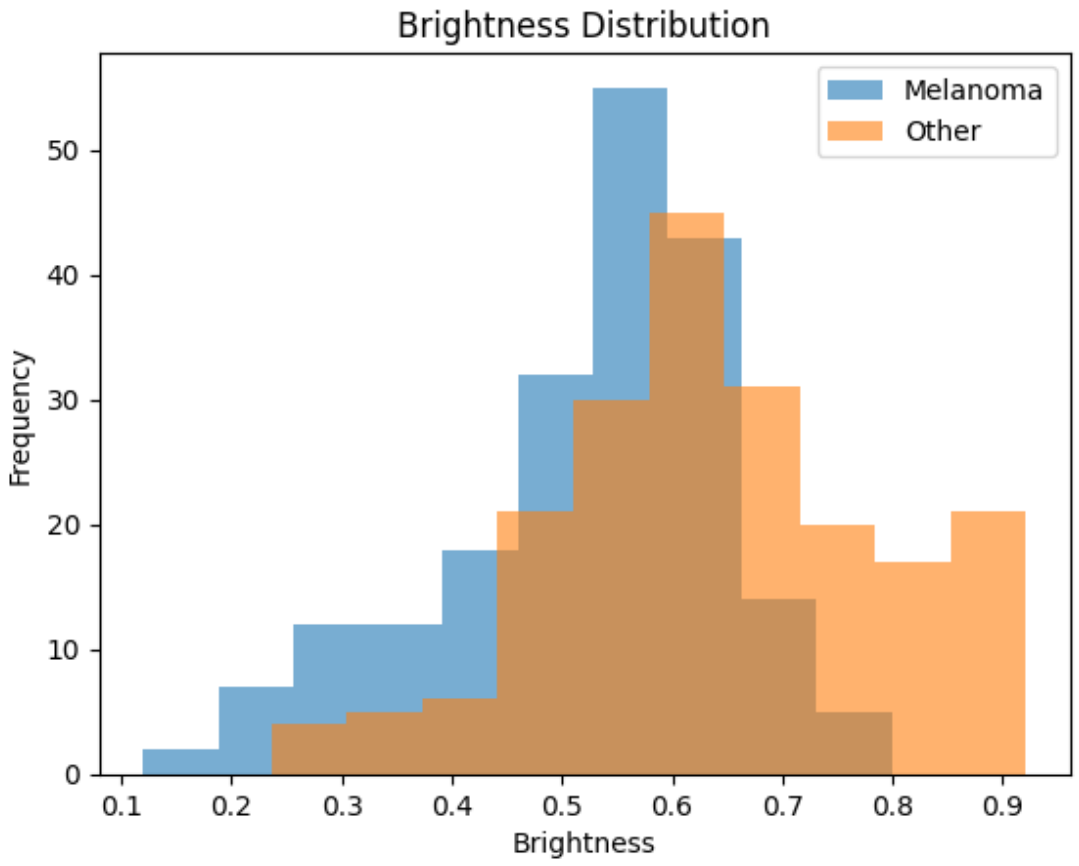
Lequan Yu et al, Automated Melanoma Recognition in Dermoscopy Images via Very Deep Residual Networks



SAMPLE IMAGES: MELANOMA VS OTHER

DATA & KEY INSIGHTS





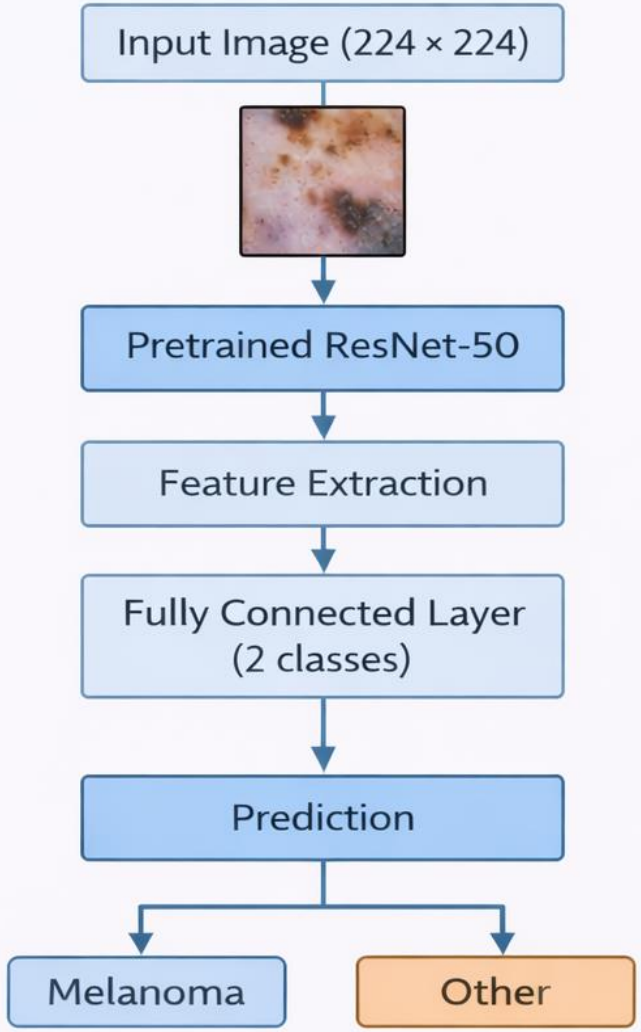
DATASET OVERVIEW

- ISIC dermoscopic image dataset for melanoma classification
- Binary classification: Melanoma vs Other
- Train/Validation/Test split used for robust evaluation

DATA QUALITY INSIGHT

- Variations in image brightness and illumination across samples
- Potential impact on model generalisation
- Normalisation applied during preprocessing to standardise inputs

MODEL APPROACH



PREPROCESSING

- Images resized to 224 × 224
- Normalisation using ImageNet mean and standard deviation
- Augmentations: flipping, rotation, colour jitter

MODEL ARCHITECTURE

- ResNet-50 pretrained on ImageNet
- Final fully connected layer adapted for binary classification
- Transfer learning used to leverage learned visual features

TRAINING STRATEGY

- CrossEntropyLoss for classification
- Adam optimizer with learning rate 1 × 10 − 4
- Learning rate scheduling (ReduceLROnPlateau)
- Data augmentation applied to improve generalisation

MODEL PERFORMANCE

ACCURACY: ~68%

Performance is moderate due to dataset complexity and class variability

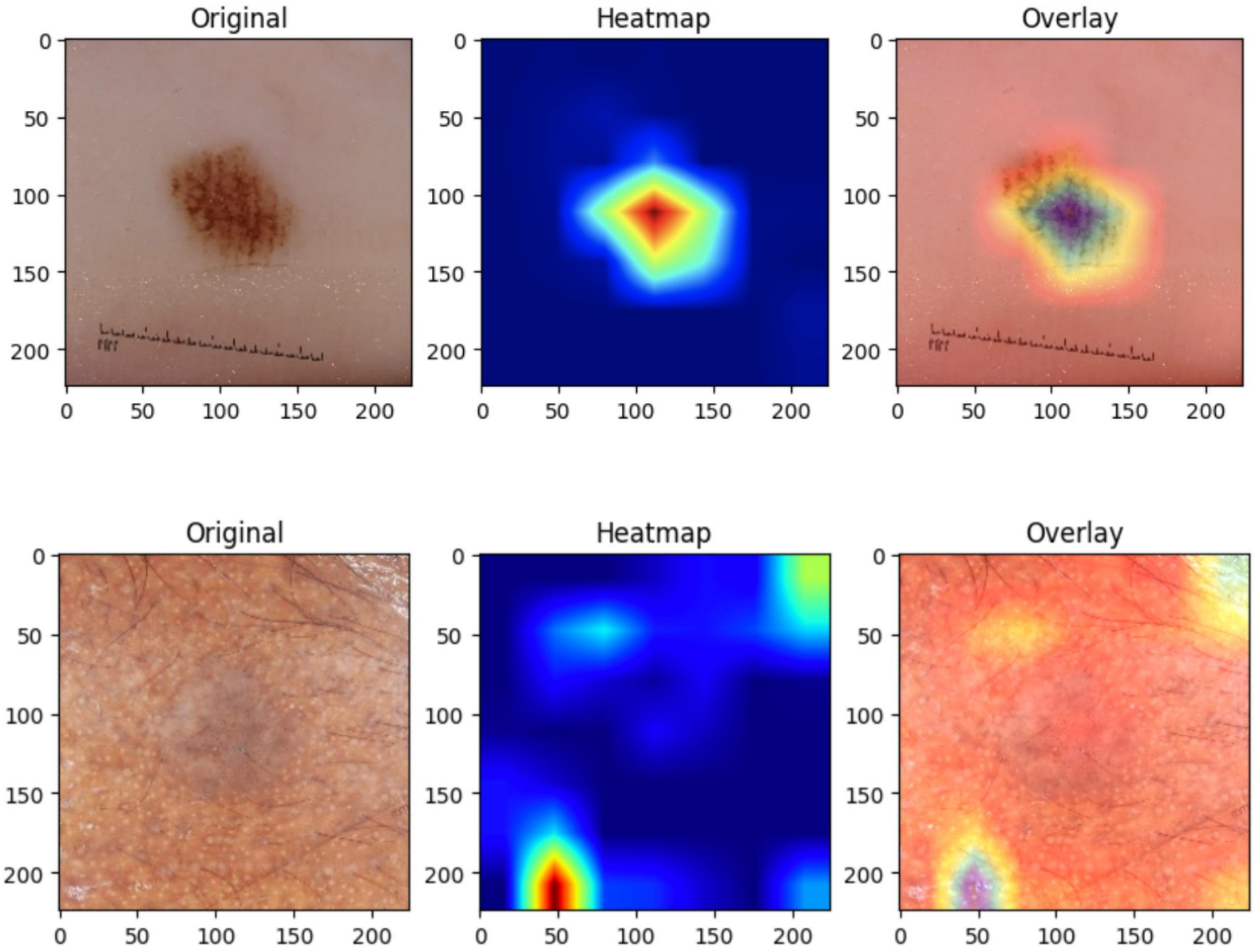
RECALL (MELANOMA): ~82%

High recall ensures most melanoma cases are correctly identified

ROC-AUC: ~0.93

Strong class discrimination ability across decision thresholds

GRAD- CAM



ANALYSIS

- Model focuses on lesion regions rather than background
- Highlights clinically relevant areas in melanoma cases
- Provides visual explanation for model predictions

INTERPRETABILITY & LIMITATIONS

LIMITATIONS

- Class imbalance affects precision
- Visual similarity between benign and malignant lesions
- Limited dataset diversity

FAILURE CASES

- Model misclassifies benign lesions as melanoma (false positives)
- Some melanoma cases missed due to subtle features
- Sensitivity to lighting and image quality

TRUST & RELIABILITY

- Grad-CAM improves transparency
- Model should assist, not replace clinicians
- Requires further validation on diverse datasets

THANK YOU